SESSION 3 — MORNING

Financial Statement Analysis I

Income statement • Earnings per share • Balance sheet analysis

Session Roadmap — 2 Hours

Understanding financial statements is the foundation of every equity, credit and M&A analysis. This session covers how they are built.

01 **FSA Framework & 3 Statements**
Why FSA • IFRS vs US GAAP • linkages between statements • what analysts seek

02 **Income Statement Deep Dive**
Revenue recognition (5 steps) • AVI long-term contract (CFA LM2) • capitalise vs expense

03 **Non-Recurring Items & EPS**
Discontinued operations • basic EPS • diluted EPS • AB InBev (CFA LM2) • CWC treasury stock

04 **Balance Sheet Structure**
Assets, liabilities, equity • intangibles • goodwill • financial instruments • deferred tax

05 **Common-Size & BS Ratios**
Apple vs Microsoft (CFA LM3) • Company A/B/Industry: quick=0.43, leverage=2.22

06 **Practice & Takeaways**
5 worked problems • formula reference card

Learning Objectives

Apply the FSA Framework

Describe the six-step FSA framework. Explain the purpose, sources and limitations of financial statement analysis.

Analyse the Income Statement

Apply the 5-step revenue recognition model. Explain capitalisation vs expensing. Compute gross, operating and net margins.

Compute Basic and Diluted EPS

Calculate basic EPS using weighted average shares. Apply if-converted and treasury stock methods for diluted EPS (CFA LM2 examples).

Read the Balance Sheet

Classify assets, liabilities and equity. Explain goodwill, intangibles, financial instruments and deferred tax liabilities.

Interpret Common-Size Statements

Perform vertical common-size analysis. Compare Apple vs Microsoft (CFA LM3 Example 1). Know CFA sector balance sheet patterns.

Calculate Balance Sheet Ratios

Compute liquidity ratios (quick=0.43 for Company A, CFA LM3) and solvency ratios (leverage=2.22 for Company B, CFA LM3).

SECTION 01

FSA Framework & The 3 Statements

The FSA Framework — 6 Steps (CFA LM1)

CFA LM1 presents a structured 6-step process for conducting financial statement analysis. Every analysis should follow this sequence.

1

Articulate Purpose & Context

Why is the analysis being done? (Equity valuation? Credit risk? M&A screening?) This determines what data to collect and what ratios matter.

2

Collect Data

Annual reports, 10-K filings, earnings releases, MD&A, auditor reports. Also external: industry data, macro data, competitor financials.

3

Process Data

Calculate ratios, produce common-size statements, adjust for one-off items, normalise earnings. Ratios are OUTPUT of this step.

4

Analyse/Interpret Data

Ratios are INPUT here. Compare over time (trend), across peers (cross-sectional). Identify drivers and risks. CFA Practice Q1: ratios are an output of Step 3, input of Step 4.

5

Develop Conclusions

Write up investment thesis, credit recommendation or due diligence memo. Communicate limitations and assumptions clearly.

6

Follow-Up

CFA LM1: follow-up involves gathering new information and repeating the analysis to determine if updates are needed.

The Three Primary Financial Statements

Three interlocking statements together describe a company's full financial picture. No single statement tells the complete story.

Income Statement (P&L)

Covers a PERIOD of time.
Revenue − Expenses = Net Income.
Shows: profitability, growth, margins.
Key metrics: EPS, EBITDA, operating margin.
Limitation: accrual basis — revenue ≠ cash received.

Balance Sheet (Statement of Financial Position)

Shows POINT IN TIME snapshot.
Assets = Liabilities + Equity.
Shows: liquidity, solvency, capital structure.
Key metrics: quick ratio, D/E, net debt.
Limitation: historical cost — some assets not at fair value.

Cash Flow Statement

Covers a PERIOD of time.
Operating + Investing + Financing = Net cash change.
Shows: actual cash generation, quality of earnings.
Key metrics: FCF = CFO − Capex.
Limitation: indirect method obscures operating details.

IFRS vs US GAAP — Key Differences for Analysts (CFA LM1)

CFA LM1: IFRS is issued by the IASB; US GAAP by the FASB. Revenue recognition standards converged in 2018. Other areas still differ.

Topic	IFRS	US GAAP	Analyst Impact
Revenue recognition	IFRS 15 (converged with ASC 606 from 2018)	ASC 606 (same 5-step model)	CONVERGED — same treatment
Inventory	FIFO or weighted average (LIFO prohibited)	FIFO, LIFO or weighted average	LIFO reserve adjustment needed to compare US firms using LIFO
Development costs	Capitalise if criteria met (IAS 38)	Expense as incurred	IFRS firms show higher assets; better earnings smoothing
Investment property	Fair value model permitted (gains in P&L)	Historical cost only	IFRS firms may show more volatility from fair value changes
Goodwill impairment	Annual impairment test (no amortisation)	Annual impairment test (2-step prior; simplified)	Both impair but testing methodology differs
Financial statement	IS, BS, CFS, changes in equity	IS, BS, CFS, changes in equity	Same primary statements; notes differ

Auditor’s Report — The Seal of Approval

CFA LM1: the independent auditor provides REASONABLE assurance (not absolute certainty) that financial statements are fairly presented. Analysts must read the opinion type.

Unqualified (Clean) Opinion

Financial statements present the company fairly in ALL MATERIAL RESPECTS in accordance with applicable accounting standards. CFA LM1: this is the only acceptable opinion for most lenders and most investors. The most common opinion for listed companies.

Qualified Opinion

Except for specific described issues, statements are fairly presented. Signals that the scope of audit was limited or there are specific EXCEPTIONS to accounting standards. Analyst must evaluate the significance of the exceptions carefully.

Adverse Opinion

Financial statements do NOT present the company fairly. Material departures from accounting standards. EXTREMELY RARE for listed companies. Immediate red flag.

Disclaimer of Opinion

Auditor unable to form opinion (e.g., scope limitation, insufficient evidence). Also a serious red flag. Analyst should treat as a major information risk.

SECTION 02

Income Statement (CFA LM2)

Income Statement Structure — From Revenue to EPS

The income statement moves from top-line revenue to bottom-line net income through a series of deductions. Each line reveals a different profitability measure.

Line Item	Meaning	Key Ratio
Revenue (Net Sales)	All goods/services sold, net of returns and discounts	Revenue growth rate
Cost of Goods Sold (COGS)	Direct production costs (materials, direct labour, manufacturing overhead)	Gross margin = Gross profit / Revenue
Gross Profit	Revenue − COGS. Measures production profitability.	Gross margin %
Operating Expenses (SG&A, R&D, D&A)	Overhead, selling, admin, depreciation	Operating income / Revenue = EBIT margin
EBIT (Operating Income)	Earnings Before Interest and Tax. Core recurring business profit.	EBIT margin %
Interest Expense	Cost of debt. Reduces pre-tax income.	Interest coverage = EBIT / Interest
EBT (Pre-Tax Income)	EBIT minus interest. Subject to tax.	Pre-tax margin %
Income Tax Expense	Effective tax rate × EBT. Note: deferred vs current tax.	Effective tax rate %
Net Income	Bottom line. Available to shareholders after all deductions.	Net margin % • ROE • EPS

Revenue Recognition — IFRS 15 / ASC 606 (CFA LM2)

Core principle: revenue should be recognised to depict the transfer of promised goods or services to customers

As of 2018, IFRS 15 and ASC 606 converged to a unified 5-step model (CFA LM2). This replaced many industry-specific rules.

Step 1

Identify the contract

Written or verbal. Must have commercial substance, approval by both parties, and collection is probable.

Step 2

Identify performance obligations

Distinct promises to transfer goods or services to the customer. A bundle may need to be broken into separate obligations.

Step 3

Determine transaction price

Total consideration expected. Variable consideration (rebates, refunds) estimated at expected value or most likely amount.

Step 4

Allocate transaction price

Allocate to each performance obligation based on relative STANDALONE selling prices. Not necessarily the contract price ratio.

Step 5

Recognise revenue

When (or as) each performance obligation is SATISFIED — i.e., control transfers to the customer. Can be point-in-time or over time.

Revenue Recognition: Long-Term Contracts — CFA LM2 (AVI)

Armored Vehicles Inc. (AVI) builds weapons systems on multi-year government contracts. Revenue is recognised over time as performance obligations are satisfied.

CFA LM2 — ARMORED VEHICLES INC (AVI) LONG-TERM CONTRACT

Contract: USD 10 million total price. Expected total cost: USD 7 million. Expected profit: USD 3 million. Duration: 2 years.

Year 1 actual: AVI incurs USD 4.2 million of costs = 60% of total estimated costs.
Revenue recognised Year 1 = 60% × USD 10m = USD 6 million
Profit recognised Year 1 = 60% × USD 3m = USD 1.8 million
Costs recognised Year 1 = USD 4.2 million

Year 2: system completed. Actual total cumulative costs = USD 7.5 million (over budget by 0.5m).
Revenue Year 2 = USD 10m − USD 6m = USD 4 million
Costs Year 2 = USD 7.5m − USD 4.2m = USD 3.3 million
Profit Year 2 = USD 4m − USD 3.3m = USD 0.7 million (cumulative profit = USD 2.5m, not 3m)

Analyst lesson: for percentage-of-completion contracts, cost overruns REDUCE profit. Track progress and estimate-at-completion carefully.

CFA requires: performance obligation satisfied over time if customer simultaneously receives and consumes benefits, or if entity has right to payment for work completed to date.

Capitalisation vs Expensing — Impact on Financial Statements (CFA LM2)

Capitalise = put on balance sheet as asset, then amortise over time to income statement

Expense = charge immediately

Impact	Capitalise in Year 1	Expense in Year 1	Long-Run Effect
Income statement	Higher NI (lower expense in year 1)	Lower NI (full expense year 1)	Cumulative NI same; timing differs
Balance sheet	Higher assets, higher equity	Lower assets, lower equity	No permanent difference
CFO vs CFF	Capitalised costs as CFI outflow (not CFO)	Expensed costs reduce CFO	CFO appears higher under capitalise
Earnings quality	Lower quality: delays expense recognition	Higher quality: more conservative	Expense → more conservative accounting

IFRS: Development Costs

Under IFRS (IAS 38): capitalise if technical feasibility, intention to complete, ability to use/sell, and future economic benefits are all demonstrable.
Research costs: always expense.
US GAAP: expense all R&D (except certain software development costs)

Interest Capitalisation

Both IFRS and US GAAP allow (require under US GAAP) capitalisation of interest costs incurred while constructing long-lived assets.
Capitalised interest increases asset cost and reduces interest expense on IS. Add back to get true interest burden.

Depreciation Methods — Impact on Reported Earnings

Depreciation method affects the TIMING of expense recognition but NOT the total amount. Total depreciation over the asset's life is always the same.

Method	Description	Year 1 Expense (USD200k asset, 5yr, USD0 salvage)	Relative Pattern
Straight-line	Equal charge each year	USD 40,000 per year	Constant: 40k each year
Double-declining balance	2/life × Book value (accelerated)	2/5 × 200k = USD 80,000	Higher early, lower later
Units of production	Expense per unit produced, varying	Depends on output	Follows production volume
Sum of years digits	Fractional: (n remaining)/(sum)	Year 1: 5/15 × 200k = USD 66,667	Accelerated, smoother than

Analyst implications of depreciation policy

Accelerated depreciation: lower reported NI early, lower taxes paid early (tax benefit from accelerated write-off). Creates deferred tax liability on balance sheet.

Analysts adjusting across companies: recalculate using consistent method. EBITDA strips out D&A entirely to improve comparability.

Asset age analysis: accumulated depreciation / gross PP&E = proportion of life used. High ratio = ageing asset base needing capex reinvestment.

Non-Recurring Items — Separating Sustainable from One-Off Earnings

Analysts must distinguish RECURRING earnings (basis for valuation multiples) from ONE-OFF items (cannot be capitalised in DCF or P/E). CFA LM2 covers three types.

Item Type	Definition	Treatment on IS	Analyst Action
Unusual/infrequent	Items that are unusual in nature OR infrequent in occurrence. BUT still from continuing operations.	Reported within continuing operations (above operating line or below). CFA: fire damage is unusual but still continuing operations.	Add back to find normalised EBIT. Include in DCF as non-recurring.
Discontinued operations	A decision to dispose of or shut down an entire operating division.	Reported BELOW the line from continuing operations: 'Income from discontinuedoperations, net of tax'.	Exclude from valuation of continuing business. Value separately.
Accounting policy changes	Change in method (e.g. FIFO to weighted average). Changes in estimate (e.g. useful life).	Retrospective application: restate ALL prior periods presented. Note disclosures.	Restate historical series. Verify rationale is reasonable.
Accounting error corrections	Prior period misstatements corrected in current period.	Restatement of prior period financials. Retained earnings adjusted.	Serious red flag for internal controls. Investigate cause.

Common-Size Income Statement — Margins & Benchmarking

A common-size IS expresses every line as a % of REVENUE. This removes the effect of size and enables direct comparison across companies and time periods.

Line Item	Company A	Company B	Industry Avg	Interpretation
Revenue	100%	100%	100%	Anchor
COGS	58%	72%	65%	A more efficient production. B's GM only 28% v for A.
Gross Margin	42%	28%	35%	A has pricing power or lower input costs. Investi which.
SG&A	18%	14%	17%	B controls overhead better. A spends more on marketing/admin.
D&A	5%	3%	4%	A more capital-intensive. Check asset age and ca intensity.
Operating Margin	19%	11%	14%	A significantly more profitable at operating leve
Interest Expense	1%	4%	2%	B more levered. Higher debt service eats into NI
Net Margin	13%	5%	9%	A's leverage advantage + operating margin = far NI.

Inventory Valuation Methods — FIFO, LIFO & Weighted Average (CFA LM2)

The choice of inventory cost method affects COGS, gross margin, net income, inventory on balance sheet and tax paid. CFA LM3 notes: LIFO prohibited under IFRS.

Method	COGS (Rising Prices)	Ending Inventory (BS)	Net Income	Tax Paid	Allowe
FIFO (First In, First Out)	Lower (older, cheaper units)	Higher (newer, costlier units)	Higher	Higher	IFRS an GAAP
LIFO (Last In, First Out)	Higher (newer, costlier units)	Lower (older, cheaper units)	Lower	Lower	US GAA — PRO under I
Weighted Average	Between FIFO and LIFO	Between FIFO and LIFO	Between	Between	Both IF US GAA

LIFO Reserve — Adjusting US GAAP LIFO Companies to FIFO Comparability

US companies using LIFO must disclose the LIFO Reserve: the difference between FIFO inventory value and LIFO inventory value.
To compare a LIFO firm to FIFO peers:
FIFO Inventory = LIFO Inventory + LIFO Reserve
FIFO COGS = LIFO COGS − Change in LIFO Reserve
FIFO Net Income = LIFO Net Income + Change in LIFO Reserve × (1 − tax rate)

CFA analyst note: in periods of RISING prices, LIFO understates inventory (balance sheet) and overstates COGS → lower NI → lower taxes. Adds a cash tax benefit but creates a balance sheet asset distortion.

Earnings Quality — Accruals and Red Flags for Analysts

High quality earnings are: (1) sustainable, (2) derived primarily from cash-generating operations, (3) not distorted by aggressive accounting choices.

Accruals = Net Income − CFO **High accruals relative to assets = lower earnings quality (Sloan accruals ratio)**

Red Flag	What It Suggests	How to Detect
Revenue growing faster than cash collections	Revenue recognition may be too aggressive. Building up fictitious or uncollectible AR.	AR / Revenue rising YoY. AR days (DSO) rising.
Gross margin expanding while revenue flat	Channel stuffing, bill-and-hold, or aggressive rev rec. Competitors not showing same trend.	Compare to peers; check channel inventory levels.
Frequent changes in accounting estimates	Stretching useful lives of PP&E, lowering warranty provisions, reducing bad debt allowances.	Changes in notes; compare to industry norms.
Large deferred revenue declining rapidly	SaaS companies: customer churn rising, or pulling forward subscription revenue.	Track deferred revenue balance YoY.
Net income consistently exceeds CFO	If accruals are large and positive over multiple years, future write-downs likely.	Sloan ratio = (NI − CFO) / Avg Total Assets > 5% = elevated risk.

SECTION 03

Earnings Per Share (CFA LM2)

Basic Earnings Per Share — Weighted Average Shares

Basic EPS = (Net Income − Preferred Dividends) / Weighted Average Shares Outstanding

Preferred dividends are deducted because they represent a prior claim on earnings before common shareholders. WASO weights shares by time outstanding.

EXAMPLE: Weighted Average Shares Calculation

Company has 1,000,000 shares at 1 January. Issues 300,000 new shares on 1 April (outstanding for 9/12 of year). Repurchases 200,000 shares on 1 October (outstanding for 3/12).

WASO = 1,000,000 × 12/12 + 300,000 × 9/12 − 200,000 × 3/12
= 1,000,000 + 225,000 − 50,000 = 1,175,000

Net income = USD 5,875,000. No preferred dividends.
Basic EPS = 5,875,000 / 1,175,000 = USD 5.00 per share

Stock dividends and stock splits: treated as if they occurred at the BEGINNING of the period. Retroactively adjust ALL prior periods shown. This preserves comparability.

CFA Practice Q7 (LM2)

Company has USD 1,000,000 net income. 1,050,000 WASO (adjusted for stock dividends). Basic EPS = 1,000,000 / 1,050,000 = USD 0.95. CFA answer: A.

Simple vs Complex Structure

Simple: only common stock. Basic = Diluted EPS. Complex: also has convertibles, options, warrants. Must compute BOTH basic and diluted EPS.

AB InBev EPS — CFA LM2 Exhibit 12

Directly from CFA Vol 4 LM2. Companies report BOTH basic and diluted EPS from continuing operations AND from total income.

AB InBev EPS (USD)	2017	2016	2015
Basic EPS (total)	4.06	0.72	5.05
Diluted EPS (total)	3.98	0.71	4.96
Basic EPS from continuing operations	4.04	0.69	5.05
Diluted EPS from continuing operations	3.96	0.68	4.96

CFA LM2 Analysis of AB InBev EPS

Key observation 1: 2017 basic EPS (4.06) >> 2016 (0.72). CFA asks analysts to investigate the CAUSE of the large increase.
Key observation 2: Diluted EPS (3.98) < Basic EPS (4.06). The difference = dilution from potentially convertible instruments (options, warrants, convertible bonds). Dilution is always negative or zero.
Key observation 3: 2015 basic EPS was 5.05, higher than 2017 (4.06). Despite revenue growth, 2017 EPS did not recover to 2015 level. Margins declined (as shown in profitability module).
Analyst task: reconcile changes to EPS with changes in net income AND share count.

Diluted EPS — Treasury Stock Method (CFA LM2 CWC)

For options and warrants: use the treasury stock method. Assumes the company uses the proceeds to BUY BACK shares at the average market price.

Diluted EPS = (NI − Preferred Div) / (WASO + Net New Shares from Conversion)

CFA LM2 CWC COMPANY — Treasury Stock Method (Practice Q10)

Data: Net income = USD 12,000,000. Preferred dividends = USD 800,000. WASO = 2,000,000 shares.
Stock options: 100,000 options. Exercise price = USD 20. Average market price = USD 25.

Step 1: Proceeds from option exercise = 100,000 × USD 20 = USD 2,000,000
Step 2: Shares repurchased at market price = USD 2,000,000 / USD 25 = 80,000 shares
Step 3: Net new shares = 100,000 − 80,000 = 20,000 shares (these DILUTE existing holders)
Step 4: Diluted shares = 2,000,000 + 20,000 = 2,020,000

Diluted EPS = (12,000,000 − 800,000) / 2,020,000 = 11,200,000 / 2,020,000 = USD 5.54

CFA answer: B is correct. Diluted EPS = USD 5.54 vs Basic EPS = (12m−800k)/2m = USD 5.60.

Anti-dilutive securities: if conversion RAISES EPS (exercise price > market price for options), EXCLUDE from diluted EPS. Diluted EPS must always be ≤ basic EPS.

Diluted EPS — If-Converted Method (Bonds & Preferred)

For convertible bonds and convertible preferred shares: use the if-converted method. Assume all convertible instruments were converted at the START of the period.

If-Converted: Add after-tax interest (bonds) or preferred dividends to numerator; add converted shares to denominator

EXAMPLE — Convertible Bond Diluted EPS

Net income = USD 10,000,000. WASO = 5,000,000. No preferred dividends.
Convertible bond: USD 10 million principal at 5% interest. Each USD 1,000 bond converts to 50 shares = 500,000 new shares if converted.

If-converted numerator: NI + after-tax interest saved = 10,000,000 + (10,000,000 × 5% × (1 − 30% tax)) = 10,000,000 + 350,000 = USD 10,350,000
If-converted denominator: WASO + converted shares = 5,000,000 + 500,000 = 5,500,000

Diluted EPS = 10,350,000 / 5,500,000 = USD 1.882 per share
Basic EPS = 10,000,000 / 5,000,000 = USD 2.000 per share
Dilution = USD 0.118 per share (5.9% dilution)

CFA Practice Q9: Diluted EPS = Net income / (WASO + new shares) = 200,000,000 / 54,000,000 = USD 3.70

Convertible preferred: add preferred dividends back to numerator (no longer paid if converted); add converted shares to denominator.

SECTION 04

Balance Sheet (CFA LM3)

Balance Sheet Structure — The Accounting Equation

Assets = Liabilities + Shareholders' Equity (at a SPECIFIC POINT IN TIME)

Category	Contents	Measurement	Key Analyst Focus
Current Assets	Cash, marketable securities, AR, inventory, prepaid exp. Convert to cash within 1 year.	Generally at fair value or lower of cost/NRV	Liquidity, working capital management
Non-Current Assets	PP&E, intangibles, goodwill, long-term investments, deferred tax assets.	Historical cost less accumulated depreciation/impairment	Capex intensity, goodwill impairment risk
Current Liabilities	Accounts payable, short-term debt, accruals, deferred revenue due within 1 year.	Carrying value (usually face value)	Near-term refinancing risk, payables management
Non-Current Liabilities	Long-term debt, pension obligations, deferred tax liabilities, lease obligations.	Amortised cost or fair value depending on type	Capital structure, interest coverage, debt maturity
Shareholders' Equity	Share capital + Retained earnings + AOCI + Treasury stock.	Book value (historical)	Book value per share, ROE drivers, buybacks

Intangible Assets & Goodwill — CFA LM3

CFA LM3: intangible assets are non-physical assets that generate future economic benefits. They are increasingly important for technology and pharmaceutical companies.

Identifiable Intangibles

Separately identifiable and can be sold independently.
Examples: patents, trademarks, brand names, customer lists, software, licences, franchises.

IFRS (IAS 38): capitalise if probable future benefits + reliable cost measurement.
US GAAP: most internally generated intangibles EXPENSED. Only acquired intangibles (via purchase) capitalised.

Amortised over useful life (finite). No amortisation if indefinite life (tested for impairment annually).

Goodwill

ONLY arises from an ACQUISITION.
Goodwill = Purchase price − Fair value of net identifiable assets acquired.
Reflects: synergies, brand premium, customer relationships not separately identifiable.

CFA LM3 Practice Q2: goodwill is affected by the acquisition PURCHASE PRICE, not the acquired company's book value.

NOT amortised (under IFRS and US GAAP). Tested for impairment annually. Large goodwill write-offs signal M&A value destruction.

Microsoft's goodwill = 14.6% of total assets (CFA LM3 Example 1). Apple's goodwill = only 1.5% — Apple grew organically, Microsoft via acquisitions (Nokia, LinkedIn, Activision etc.).

Financial Instruments — Three Classifications (CFA LM3)

How a financial asset is CLASSIFIED determines how it appears on the balance sheet and income statement. CFA LM3 tests all three categories.

Classification	Held For?	Balance Sheet Measurement	Unrealised Gains/Losses	CFA Answer
Trading Securities (FVTPL)	Short-term profit	Fair value	Income statement → retained earnings	CFA LM3 Q3: answer B (flow income)
Available-for-Sale (AFS/FVOCI)	Neither trading nor HTM	Fair value	OCI → Accumulated OCI in equity (NOT IS)	CFA LM3 Q4: answer C (AOC equity)
Held-to-Maturity (HTM)	Hold to maturity	Amortised cost (no fair value changes)	Not recognised (realised only)	CFA LM3 Q5: answer A (not recognised)

Why classification matters to analysts

Trading securities: fair value moves create income volatility. Good for mark-to-market transparency but may mask underlying business performance.
AFS securities: fair value moves bypass IS. Equity can be volatile without NI being affected. Watch AOCI for hidden losses.
HTM: insulates NI and equity from market moves. But only valid if company genuinely intends and is able to hold to maturity. Reclassification is rare and restricted.

Deferred Taxes — When Tax and Accounting Diverge (CFA LM3)

Temporary difference = taxable income ≠ accounting income. Creates deferred tax asset or liability.

Deferred tax arises because tax authorities allow different timing of income/deductions vs GAAP/IFRS. NOT a permanent difference — it will reverse.

Deferred Tax Liability (DTL)

Taxable income < Accounting income now.
Taxes payable now < Tax expense now.
Difference = DTL (owe more tax in future).

Common cause: accelerated depreciation for tax (larger deductions now) but straight-line on accounts.
SAP balance sheet shows EUR 240m of DTL.
Apple: USD 31,504m in DTL (within other non-current liabilities).

Deferred Tax Asset (DTA)

Taxable income > Accounting income now.
Taxes payable now > Tax expense now.
Difference = DTA (prepaid taxes).

Common causes: warranty provisions, unearned revenue recognised early for tax, carry-forward losses.
DTA only recognised if it is PROBABLE the company will have future taxable income to use it against. Impaired if not.

Effective tax rate vs statutory rate: differences explained by permanent differences (tax-exempt income, non-deductible expenses), tax credits, and deferred tax changes.

SECTION 05

Common-Size & Balance Sheet Ratios (CFA LM3)

Cross-Sectional Analysis — Apple vs Microsoft (CFA LM3 Example 1)

Directly from CFA Vol 4, LM3, Example 1. Both are IT sector peers. Common-size comparison reveals very different business models and strategies.

Balance Sheet Item (% of Total Assets)	Apple 2017	Microsoft 2017	Key Interpretation
Cash + Short-term securities	19.8%	55.2%	Microsoft holds far more liquid assets — deferred re from software subscriptions.
Long-term marketable securities	51.9%	2.5%	Apple parks its massive cash hoard in long-term secu Very unusual.
Goodwill	1.5%	14.6%	Apple: organic growth. Microsoft: acquisition-driven LinkedIn etc.).
Long-term debt	25.9%	31.6%	Both carry significant debt — tax-efficient in low-rate environment.
Total liabilities	64.3%	70.0%	Both highly leveraged vs their assets. Cash-rich but a laden.
Total shareholders' equity	35.7%	30.0%	Apple's equity slightly higher. Microsoft's intangibles equity base.

CFA observation: Apple's low accounts receivable (4.8%) vs Microsoft (8.2%) reflects Apple's use of direct retail and cash sales. Apple's high accounts payable (13.1%) = strong bargaining power with suppliers.

Balance Sheet Ratios — CFA LM3, Exhibit 13

CFA LM3, Exhibit 13 provides these ratios and their interpretation. Test your ability to calculate and compare to industry.

Ratio	Formula	What It Measures	Typical Reading
Current Ratio	Current Assets / Current Liabilities	General short-term liquidity	1.0–3.0x; < 1 = concern; >3 = inefficient use of current assets
Quick Ratio (Acid Test)	(Cash + Mkt Sec + Receivables) / Current Liabilities	Liquidity excluding inventory and prepaid	0.5–1.5x; most stringent test excluding inventory
Cash Ratio	(Cash + Mkt Securities) / Current Liabilities	Immediate cash-only liquidity	0.2–1.0x; most conservative metric
Long-term Debt to Equity	Long-term Debt / Total Equity	Long-term leverage	<1.0x generally; capital-intensive may be higher
Debt-to-Equity	Total Debt / Total Equity	Total leverage (ST + LT debt)	Varies widely by industry
Financial Leverage	Total Assets / Total Equity	Magnification effect on ROE	1.0 = no debt; 2.0 = 50% of assets funded by debt
Debt-to-Assets	Total Debt / Total Assets	What % of assets financed by creditors	< 0.5 conservative; > 0.7 aggressive

CFA LM3 Practice Q12-15 — Company A, B & Industry Comparison

Directly from CFA Vol 4, LM3 Practice Problems. Three sets of common-size balance sheet data. Tests ratios and interpretation.

Item (% of Total Assets)	Company A	Company B	Industry Avg
Cash	5	5	7
Marketable Securities	5	0	2
Accounts Receivable	5	15	12
Inventories	15	20	16
Prepaid Expenses	5	15	11
Total Current Assets	35	55	48
PP&E, net	40	35	37
Goodwill	25	0	8
Total Current Liabilities	35	25	28
Long-term Debt	45	20	28
Total Liabilities	80	55	63
Total Shareholders' Equity	[illegible]	[illegible]	[illegible]

CFA Answers

Q12 answer C: Company A has goodwill = 25% of assets → has made acquisitions (CFA: answer C is correct; B is wrong — Co B has lower debt, not higher solvency risk than industry).

Q13: Quick ratio A = (Cash + Mkt Sec + AR) / CL = (5+5+5)/35 = 15/35 = 0.43 (CFA answer A).

SAP Group Balance Sheet Ratios — CFA LM3

CFA LM3 uses SAP Group (German software company) balance sheet data to demonstrate ratio calculation and trend analysis.

Ratio	2017 (EUR million)	2016 (EUR million)	Trend
Current Ratio	11,930 / 10,210 = 1.17x	11,564 / 9,674 = 1.20x	↓ Declining liquidity
Quick Ratio (approx)	(4,011 + 990 + 5,899) / 10,210 = 1.07x	(3,702 + 1,124 + 5,924) / 9,674 = 1.11x	↓ All three liquidity ratios fell in 2017
Cash Ratio	(4,011 + 990) / 10,210 = 0.49x	(3,702 + 1,124) / 9,674 = 0.50x	↓ Slightly less cash relative to CL
Long-term Debt to Equity	5,034 / 25,540 = 19.7%	6,481 / 26,397 = 24.6%	↓ Improving solvency (less LT debt)
Debt-to-Equity (total)	(1,561+5,034) / 25,540 = 25.8%	(1,813+6,481) / 26,397 = 31.4%	↓ Leverage reduced
Financial Leverage	42,497 / 25,540 = 1.66x	44,277 / 26,397 = 1.68x	↓ Slightly less leveraged

CFA LM3 answers: Q2 (all three liquidity ratios fell in 2017) = A, B, C all correct. Q3 (all three leverage ratios fell in 2017) = A, B, C all correct.

PRACTICE PROBLEMS

Work through each before checking answers.

Q1	AVI long-term contract (CFA LM2): USD 10m contract. In Year 1, AVI incurs USD 2.1m costs out of expected USD 7m total. How much revenue and profit does AVI recognise in Year 1?
Q2	Weighted average shares: Company has 4,000,000 shares on 1 January. Issues 1,200,000 shares on 1 July. Repurchases 400,000 shares on 1 October. Net income = USD 2,800,000. Calculate basic EPS.
Q3	Treasury stock method (CFA LM2 CWC): 2,000,000 WASO. NI=USD12m. Preferred div=USD800k. 100,000 options with exercise price USD20 and average market price USD25. Calculate diluted EPS.
Q4	Company A balance sheet (CFA LM3): cash=5, mkt sec=5, AR=5, prepaid=15, inventory=15, current liabilities=35 (all as % of total assets). Calculate quick ratio. Is it above or below industry average of 0.75?
Q5	Company B (CFA LM3): total assets=100%, total shareholders equity=45%, total current liabilities=25%, long-term debt=20%. Calculate (a) financial leverage ratio, (b) debt-to-equity ratio.

Practice — Worked Solutions

A1

Year 1 % completion = 2.1 / 7.0 = 30%. Revenue = 30% × USD 10m = USD 3m. Profit = 30% × USD 3m (expected) = USD 0.9m. Costs recognised = USD 2.1m (actual). P&L: Revenue 3m, Costs 2.1m, Profit 0.9m.

A2

WASO = 4,000,000 × 12/12 + 1,200,000 × 6/12 − 400,000 × 3/12 = 4,000,000 + 600,000 − 100,000 = 4,500,000. Basic EPS = 2,800,000 / 4,500,000 = USD 0.622 per share.

A3

Proceeds = 100,000 × 20 = 2,000,000. Repurchased = 2,000,000 / 25 = 80,000 shares. Net new shares = 100,000 − 80,000 = 20,000. Diluted shares = 2,020,000. Diluted EPS = (12,000,000 − 800,000) / 2,020,000 = USD 5.54. CFA answer B confirmed.

A4

Quick ratio A = (Cash + Mkt Sec + AR) / CL = (5 + 5 + 5) / 35 = 15/35 = 0.43x. Industry average quick = (7+2+12)/28 = 21/28 = 0.75x. Company A quick (0.43) is BELOW industry average. Despite having adequate current assets overall (current ratio 35/35 = 1.0x), its near-term liquidity is below peers. CFA answer A for Practice Q13.

A5

(a) Financial leverage = Total assets / Equity = 100 / 45 = 2.22x (CFA answer C for Practice Q14). (b) Debt-to-equity = Total debt / Equity = (CL + LT debt) / Equity = (25 + 20) / 45 = 45/45 = 1.0x. Total liabilities to equity = 55/45 = 1.22x.

KEY TAKEAWAYS

Session 5 — Financial Statement Analysis I

- Revenue recognition 5-step model (IFRS 15/ASC 606): identify contract, POs, price; allocate; recognise when control transfers.
- AVI (CFA LM2): Year 1 revenue = % complete × contract price. Cost overruns in Year 2 reduce total profit. Progress-based recognition.
- AB InBev (CFA LM2): basic EPS=4.06, diluted=3.98 (2017). Diluted always ≤ basic. Investigate causes of YoY EPS changes.
- CWC (CFA LM2): Treasury stock method. 100k options, exercise=20, market=25. Net new shares=20k. Diluted EPS=USD5.54.
- Apple vs Microsoft (CFA LM3): Apple: low goodwill (organic growth), high AP (supplier power). Microsoft: high goodwill (acquisitions).
- Company A quick ratio (CFA LM3) = 0.43x (below industry 0.75x). Company B leverage = 2.22x. Goodwill=25% on A's BS → made acquisitions.